

Loong Kuan Lee

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Education

- 2019–2023 **Doctor of Philosophy (PhD)**, *Monash University*, Melbourne
- 2015–2018 **Bachelor of Informatics and Computation Advanced (Honours)**, *Monash University*, Melbourne, Final Mark - 88/100
Specialised in Computer Science and Statistics & Probability. Graduated with First Class Honours.

Doctoral Thesis

- title *Computing Divergences between High Dimensional Graphical Models*
- supervisors Geoff Webb, Daniel Schmidt, Nico Piatkowski
- year 2023
- description We develop a method for computing the joint, marginal, and conditional $\alpha\beta$ - divergences, a family of divergences that include the Kullback-Leibler divergence and the Hellinger distance. We then apply this method to modifying the parameters of a decomposable model such that the resulting model is some target amount of divergence away from the original.

Honours thesis

- title *Generating Concept Drift by Shuffling Instances*
- supervisors Geoff Webb
- year 2018
- description We propose a method for shuffling the instances in a dataset such that the divergence between the empirical distribution of the first and second half of the resulting dataset reaches some target amount of divergence.

Experience

Academia

- 2016–2017 **Undergraduate Research Assistant**, *Monash University*, Melbourne
Researched **Concept Drift** , specifically how to measure and visualise concept drift in both streaming and static data.
Tasks:
 - Developed a system to incrementally measure changes to the probability distributions of a data set over time, using **Java** and the library **Weka** .
 - Developed a companion web application for the system above using **Scala** with the **Play Framework** , **Javascript** , and **HTML** .
 - Used **R** extensively to visualise results and produce reports.

Industry

- 2016 **Winter Research Project**, *Agilent & Monash University*, Melbourne
Developed application to compare and analyse large groups of timeseries data over the same domain, using **R** and **Shiny** .
- 2017 **Software Tester**, *Carsales*, Melbourne
Tested backend APIs being developed to move product from a monolith to a microservice architecture. Helped develop a prototype model to predict car depreciation.
Tasks:
 - Created automated tests for APIs in CI pipeline using **Postman** , **Node.js** , and **Jenkins** .
 - Carried out performance testing of APIs using **Scala** and **Gatling** .
 - Managed communication across multiple teams to track bugs reported by consumers of the backend APIs and bugs I found in systems the APIs depend on.
 - Developed prototype to predict the depreciation of a car using **R** , **Azure ML Studio** , and **Vue.js** .

Scholarship & Awards

- 2015 Faculty of IT International Merit Scholarship
 2015 Dean's Achievement Award
 2016 Summer Research Scholarship — Faculty of IT
 2016 Winter Research Scholarship — Faculty of IT
 2017 Information Technology IBL (Industry Based Learning) Placement Scholarship
 2017 Dean's Achievement Award
 2019 Australian Government RTP (Research Training Program) Scholarship

Publications

Authored

Loong Kuan Lee, Thore Gerlach, and Nico Piatkowski. Standardization of Multi-Objective QUBOs. In *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*, volume 01, pages 58–64, August 2025 .

Loong Kuan Lee, Thore Gerlach, Johannes Knaute, Florian Gerhardt, Patrick Völker, Tomislav Maras, Alexander Dotterweich, and Nico Piatkowski. Multi-objective quantum power system redispatch. In *2025 IEEE 12th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 1–12. IEEE, October 2025 .

Loong Kuan Lee, Geoffrey I. Webb, Daniel F. Schmidt, and Nico Piatkowski. Computing marginal and conditional divergences between decomposable models with applications in quantum computing and earth observation. *Knowledge and Information Systems*, 66(12):7527–7556, August 2024 .

Loong Kuan Lee, Nico Piatkowski, François Petitjean, and Geoffrey I. Webb. Computing divergences between discrete decomposable models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(10):12243–12251, June 2023 .

Loong Kuan Lee, Geoffrey I. Webb, Daniel F. Schmidt, and Nico Piatkowski. Computing Marginal and Conditional Divergences between Decomposable Models

with Applications. In *2023 IEEE International Conference on Data Mining (ICDM)*, pages 239–248, December 2023 .

Coauthored

Sascha Mücke, Raoul Heese, Thore Gerlach, David Biesner, Loong Kuan Lee, and Nico Piatkowski. Quantum adiabatic generation of human-like passwords. In *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 1516–1524. IEEE, August 2025 .

Thore Gerlach, Loong Kuan Lee, Frederic Barbaresco, and Nico Piatkowski. Hybrid quantum-classical multi-agent pathfinding. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 19161–19171. PMLR, July 2025 .

Geoffrey I. Webb, Loong Kuan Lee, Bart Goethals, and François Petitjean. Analyzing concept drift and shift from sample data. *Data Mining and Knowledge Discovery*, 32(5):1179–1199, September 2018 .